Part II — Motion and Forces · Chapter 6

Describing Motion

Learning the precise language physicists use to say where things are, how they move, and relative to what

> *A bird flies across a field. A car merges onto a highway. A raindrop falls from a cloud. In each case, something is moving — but what does 'moving' actually mean? Moving relative to what? How do we record where something is, where it was, and how its position changed? Before calculating speed or force, physics needs a precise language of description. This chapter builds that language.*

6.1 What Motion Means

An object is in motion when its position changes relative to a reference point over time. This definition seems simple, but it contains two important ideas: motion always involves a change in position, and that change is always measured relative to something else.

Motion is not an intrinsic property of an object. A passenger sitting in an airplane seat is motionless relative to the airplane, but moving at 900 km/h relative to the ground, and moving at a different speed still relative to the Sun. None of these descriptions is more 'correct' than the others; they simply use different reference points. Physics accepts all of them as valid and provides the tools to translate between them.

The study of motion without asking what causes it is called kinematics. Kinematics describes the where, when, and how-fast of motion using position, time, velocity, and acceleration. The question of why objects move — what forces produce the motion — belongs to dynamics, which begins in Chapter 10. Chapters 6 through 9 are entirely kinematic: we describe motion without yet explaining it.

> **Key Point:** Kinematics asks: Where is it? How fast is it moving? How is the motion changing? Dynamics asks: Why is it moving that way? What force caused that? Keep these two questions clearly separated as you work through Part II.

6.2 Position and Change in Position

Position is the location of an object measured relative to a chosen reference point called the origin, using a coordinate system. In one dimension, position is a single signed number with a unit: x = +3.5 m means the object is 3.5 meters in the positive direction from the origin. x = −2.0 m means the object is 2.0 meters in the negative direction.

Position is not the same as distance from the origin. Position is a signed quantity; it carries direction information. Distance from the origin is the magnitude of position and is always positive.

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Position:   x = +3.5 m      (3.5 m in the positive direction)

Position:   x = −2.0 m      (2.0 m in the negative direction)
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When an object moves, its position changes. The change in position is called displacement. Displacement is calculated by subtracting the initial position from the final position:

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Δx = x_f − x_i
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Displacement is final position minus initial position. It is a signed quantity with direction.

Example — Position change A dog starts at x_i = +2.0 m and moves to x_f = +7.0 m. Displacement = 7.0 − 2.0 = +5.0 m (moved in the positive direction). A cat starts at x_i = +5.0 m and moves to x_f = +1.0 m. Displacement = 1.0 − 5.0 = −4.0 m (moved in the negative direction).

6.3 Distance and Displacement

Distance and displacement sound similar but are fundamentally different quantities. Distance is the total length of the path traveled, regardless of direction. It is a scalar: always positive, always accumulating. Displacement is the straight-line change in position from start to finish. It is a vector: it has magnitude and direction, and it can be positive, negative, or zero.

Property	Distance	Displacement
Type	Scalar (magnitude only)	Vector (magnitude and direction)
Symbol	d	₩u0394x (or ₩u0394r in 2D)
Sign	Always positive	Can be positive, negative, or zero

Path-dependent?	Yes — depends on the route taken	No — only start and end points matter
Example	A car drives 3 km north then 3 km south: distance = 6 km	Same car: displacement = 0 (returned to start)

Table 6.1 *Distance versus displacement: key differences.*

The most important distinction: if an object returns to its starting point, its displacement is zero — regardless of how far it traveled. A runner completing one lap of a 400 m track has traveled a distance of 400 m but has a displacement of zero.

Common Mistake: Distance and displacement are equal only when the object moves in a straight line without reversing direction. In all other cases, distance ≥ |displacement|.

Example — Running a lap An athlete runs one complete 400 m lap. Distance = 400 m. Displacement = 0 m. Average speed = 400 m ÷ time. Average velocity = 0 ÷ time = 0 m/s. These distinctions matter — speed and velocity give different results.

6.4 Reference Points and Reference Frames

A reference point (or origin) is the chosen location assigned coordinate zero. All positions are measured relative to this point. The choice of reference point is arbitrary; the physics does not change with the choice, but the numerical values of positions do.

A reference frame is a broader concept: it includes the reference point plus the coordinate axes and, crucially, the state of motion of the observer. Two observers in different states of motion — one standing still, one moving in a car — describe the same event with different numbers. Both descriptions are equally valid; they are simply expressed in different reference frames.

In everyday life, we almost always use Earth as our reference frame without thinking about it. In physics, we must be explicit about the reference frame whenever the motion of the observer matters.

Historical Note: Galileo Galilei was the first to state clearly that the laws of motion look the same in any reference frame moving at constant velocity. A cannonball fired on a moving ship follows the same arc (relative to the ship) as one fired on land. This insight — called the Galilean principle of relativity — is the conceptual ancestor of Einstein's theories of relativity.

Example — Reference frame matters A person walks at 2 m/s toward the front of a train moving at 30 m/s. Relative to the train: the person moves at 2 m/s forward. Relative to the ground: the person moves at 32 m/s forward. Both are correct — in their respective reference frames.

The Rule: Always state your reference point and positive direction at the start of a problem. Write it in your diagram. Every position and displacement you calculate depends on this choice.

6.5 One-Dimensional Motion

Most of Part II focuses on one-dimensional (1D) motion: motion along a single straight line. This is the simplest case and provides the foundation for understanding more complex motion in two and three dimensions.

In 1D motion, we need only one coordinate (usually called x for horizontal motion, or y for vertical motion) to specify position completely. The coordinate increases in the chosen positive direction and decreases in the negative direction. Because we are on a single axis, direction is fully encoded in the sign of the number.

Examples of 1D motion: a car driving along a straight highway; a ball thrown straight up and falling back down; an elevator moving between floors; a train moving along a straight track. In all of these, one coordinate is sufficient to describe the complete motion.

Key Point: One-dimensional motion is not just a simplification for textbooks. Many real physical situations are accurately described in 1D: free fall, motion on a straight track, compression of a spring, current in a wire. Mastering 1D kinematics gives you tools you will use throughout the book.

6.6 Motion Along a Straight Line

For motion along a straight line, we set up a number line aligned with the direction of motion. We choose an origin and a positive direction, then assign a numerical position to every point on the line.

The object's position at any moment is a single number on this line. As time passes, that number changes. The sequence of positions over time completely describes the motion.

We often record positions in a data table, with columns for time and position. Each row is a snapshot of the object at one instant. Reading down the position column tells us whether the object moved in the positive direction (position increasing), the negative direction (position decreasing), or not at all (position constant).

Time (s)	Position (m)	What Is Happening

0.0	0.0	Object at origin; at rest or about to move
1.0	+2.0	Moved 2.0 m in positive direction
2.0	+4.0	Moved another 2.0 m; equal steps — constant speed
3.0	+4.0	Position unchanged; object stopped momentarily
4.0	+1.5	Moved 2.5 m in negative direction; reversed
5.0	–1.0	Continued moving in negative direction

Table 6.2 *A position-time data table for an object moving along a straight line. Reading the position column reveals the nature of the motion.*

6.7 Positive and Negative Direction

The positive direction is the direction in which the coordinate axis increases. The negative direction is opposite. The choice of which direction is positive is made by the problem-solver, not by nature. However, once you make the choice, you must be consistent throughout the entire problem.

Common conventions: for horizontal motion, rightward is usually positive. For vertical motion, upward is usually positive (though some problems use downward as positive when an object falls). For a car on a road, the initial direction of motion is often chosen as positive.

The sign of a displacement tells you direction: positive means the object moved in the positive direction; negative means it moved in the negative direction. The sign of a position tells you which side of the origin the object is on: positive positions are on the positive side; negative positions are on the negative side.

Common Mistake: Negative position does not mean the object moved backward. An object at $x = -3$ m has simply been defined to be on the negative side of the origin. Whether this is ‘backward’ depends on the context. The sign only tells you location relative to the origin, not the history of the motion.

Tip: When you choose a positive direction, mark it clearly on your diagram with a labeled arrow. Write ‘+x →’ or ‘positive direction ↑’. This single habit eliminates most sign errors in kinematics problems.

6.8 Time Intervals and Elapsed Time

Time in physics is measured from a reference moment called $t = 0$, which is chosen by the problem-solver. It is usually the moment the motion begins or the moment an interesting event occurs, but any convenient reference point works.

A time interval (or elapsed time) is the duration between two moments: $\Delta t = t_f - t_i$. Like displacement, a time interval is a change: final value minus initial value. Time intervals are always positive (or zero) because time does not run backward in classical physics.

$$\Delta t = t_f - t_i$$

Elapsed time is always the final time minus the initial time.

Do not confuse the instant of time t with the time interval Δt. The instant $t = 3$ s tells you when something happened. The interval $\Delta t = 3$ s tells you how long a process lasted. These are different quantities and appear in different places in kinematic equations.

Example — Time interval A race starts at $t = 0$. A runner crosses the finish line at $t = 48.3$ s. The elapsed time (time interval) is $\Delta t = 48.3 - 0 = 48.3$ s. If a second runner crosses at $t = 50.1$ s, the elapsed time for the second runner is 50.1 s; the time interval between the two finishes is $50.1 - 48.3 = 1.8$ s.

6.9 Motion Diagrams

A motion diagram is a visual representation of an object's position at equally spaced time intervals. It shows a sequence of dots (each representing the object's position at one instant) along the object's path. Motion diagrams give an immediate visual sense of whether the object is moving fast or slow, speeding up or slowing down, and in what direction.

Reading a Motion Diagram

- Dots close together: the object moved a short distance between time steps — it was moving slowly.
- Dots far apart: the object moved a large distance between time steps — it was moving quickly.
- Equally spaced dots: the object moved the same distance each time step — constant speed (uniform motion).
- Increasing gaps between dots: the object moved farther each time step — it was speeding up.
- Decreasing gaps between dots: the object moved a shorter distance each time step — it was slowing down.
- Dots that reverse direction: the object turned around at the point where the spacing went to zero.

Dot Pattern (left to right)	Motion Description
• • • • • • (equal spacing)	Constant speed in positive direction
• • •• ••• •••• (increasing gaps)	Speeding up (accelerating)
•••• ••• •• • (decreasing gaps)	Slowing down (decelerating)
• • • (no dots) (at rest)	Uniform motion then stopped

Table 6.3 *Interpreting motion diagram dot patterns.*

Tip: Motion diagrams are a thinking tool, not just a drawing exercise. Before solving any kinematics problem, sketch a motion diagram. It will immediately tell you whether the object is speeding up, slowing down, or moving at constant speed, and will catch sign errors before you start calculating.

6.10 Position-Time Graphs

A position-time (x-t) graph plots an object's position on the vertical axis and time on the horizontal axis. It is one of the most information-rich representations of motion available, because its shape, slope, and features each carry physical meaning.

What the Shape of an x-t Graph Tells You

- Horizontal line (slope = 0): the object is at rest. Position is not changing.
- Straight line with positive slope: the object is moving in the positive direction at constant speed.
- Straight line with negative slope: the object is moving in the negative direction at constant speed.
- Curved line, curving upward (slope increasing): the object is speeding up in the positive direction.
- Curved line, curving downward (slope decreasing): the object is slowing down.
- A curve that goes up then comes back down: the object moved away then returned.

The Slope of an x-t Graph

The slope of a position-time graph equals the velocity of the object. A steeper slope means faster motion. A positive slope means motion in the positive direction. A negative slope means motion in the negative direction. A slope of zero means the object is at rest.

$$\text{slope of x-t graph} = \Delta x / \Delta t = \text{velocity}$$

This is the definition of velocity — covered fully in Chapter 7. For now, recognize that slope encodes speed and direction.

The x-intercept of a position-time graph gives the object's position at t = 0. The y-intercept (where the graph crosses the position axis) gives the initial position x_0.

> **Common Mistake:** A steep x-t graph means fast motion, not acceleration. Acceleration appears as curvature on an x-t graph, not steepness. A perfectly straight but steep line means constant (fast) velocity — no acceleration at all.

6.11 Uniform Motion vs. Changing Motion

Uniform motion means the object moves with constant velocity: same speed, same direction, the entire time. On an x-t graph, uniform motion produces a perfectly straight line. On a motion diagram, it produces equally spaced dots.

Changing motion means the object's velocity is changing: it is speeding up, slowing down, or changing direction. On an x-t graph, changing motion produces a curve. On a motion diagram, it produces unequally spaced dots.

Feature	Uniform Motion	Changing Motion
x-t graph shape	Straight line	Curve (parabola for constant acceleration)
Motion diagram	Equally spaced dots	Unequally spaced dots
Velocity	Constant	Changing
Acceleration	Zero	Non-zero
Physical example	Car cruising on a highway	Car accelerating from a stop; falling ball

Table 6.4 *Uniform versus changing motion — how each appears in graphs and diagrams.*

Uniform motion is the exception in the real world; most motion involves changing velocity. But understanding uniform motion first builds the conceptual foundation for understanding what acceleration means (Chapter 7) and what causes it (Chapter 10).

> **Bridge Note** *In Volume I of the advanced series, uniform motion is described as geodesic motion: an object moving without any net force follows the straightest possible path through spacetime. This is the geometric interpretation of Newton's first law.*

6.12 Describing Motion in Words, Tables, Graphs, and Diagrams

Physics uses four complementary representations to describe motion, and fluency in all four — and the ability to translate between them — is a core skill for this entire book.

Representation	Strengths	Best For
Words	Qualitative; accessible; good for summaries	Describing the general nature of motion; stating assumptions
Data table	Precise numerical values; easy to record in lab	Recording measurements; identifying specific positions and times
Position-time graph	Visual; shows trend; slope = velocity	Seeing overall motion pattern; comparing multiple objects
Motion diagram	Intuitive; shows speed changes visually	Quick sketching; identifying acceleration without calculation

Table 6.5 *The four representations of motion and their strengths.*

Expert physicists move fluidly between these representations. A good problem-solving habit is to use at least two representations for any complex motion problem: draw a motion diagram to get the qualitative picture, then use a table or graph for quantitative analysis.

> **Example — Translating representations** Words: 'A car starts at rest, accelerates for 5 seconds, then drives at constant speed.' → Motion diagram: dots with increasing gaps for 5 s, then equally spaced dots. → x-t graph: a curve (parabola) for the first 5 s, then a straight line. → Table: position increases slowly at first, then more rapidly, then at a constant rate per second.

6.13 Common Mistakes When Describing Motion

Mistake 1: Confusing distance with displacement

An object that travels 10 m to the right and 10 m back to the left has a distance of 20 m and a displacement of 0 m. Many students report the distance when the problem asks for displacement, or vice versa. Read the question carefully and identify which quantity is requested.

Mistake 2: Forgetting to state the reference point and positive direction

Without a reference point, position has no meaning. Without a stated positive direction, sign errors are almost guaranteed. Always draw and label your coordinate system before writing any numbers.

Mistake 3: Treating a steep x-t graph as acceleration

A steep straight line on an x-t graph means fast motion — not acceleration. Acceleration appears as curvature. A perfectly straight line, no matter how steep, represents constant velocity (zero acceleration).

Mistake 4: Treating time intervals and instants of time as interchangeable

The time $t = 5$ s tells you when. The time interval $\Delta t = 5$ s tells you how long. These appear in different parts of the kinematic equations and have different physical meanings.

Mistake 5: Saying an object 'has no motion' without specifying the reference frame

An object that is stationary relative to a car is moving relative to the road. Motion is always relative to a reference frame. When you say something is 'at rest,' always add 'relative to [something].'

Mistake 6: Assuming motion diagrams show a path in space

A motion diagram shows positions along the direction of motion at successive time intervals. It is not a spatial map. Dots close together mean slow motion, not a sharp turn. Always label the time interval between dots.

Chapter Summary

Motion is a change in position relative to a reference point over time. Physics describes motion using four complementary tools: words, tables, graphs, and diagrams. The key quantities are position, displacement, distance, time, and time intervals.

- Position (x) is a signed number measured from a chosen origin. It requires a reference point, a positive direction, and a unit.
- Displacement ($\Delta x = x_f - x_i$) is the vector change in position: signed, direction-dependent, and path-independent.
- Distance is the total path length traveled: a scalar, always positive, and path-dependent.
- A reference frame includes a reference point and the state of motion of the observer. Motion descriptions are always relative to a reference frame.
- Time intervals ($\Delta t = t_f - t_i$) measure duration; they are distinct from the time coordinate t, which marks a specific instant.
- Motion diagrams show dot spacing: equal = uniform motion; increasing gaps = speeding up; decreasing gaps = slowing down.

- On an x-t graph: straight line = constant velocity; curve = changing velocity; slope = velocity; steepness ≠ acceleration.
- Uniform motion: constant velocity, straight x-t graph, equally spaced dots. Changing motion: varying velocity, curved x-t graph, unequal dot spacing.

Key Terms

kinematics: The branch of physics that describes motion without reference to its causes; concerned with position, velocity, and acceleration.

motion: A change in position relative to a reference point over time.

position: The location of an object measured relative to a chosen origin using a coordinate system; a signed quantity with direction.

origin: The reference point assigned coordinate zero in a coordinate system.

displacement: The vector change in position from start to finish: $\Delta x = x_f - x_i$; signed, can be negative, path-independent.

distance: The total length of the path traveled; a scalar, always positive, path-dependent.

reference point: The chosen location assigned position zero; all positions are measured relative to it.

reference frame: A coordinate system plus the observer's state of motion; defines the perspective from which motion is measured.

positive direction: The direction along an axis in which the coordinate increases; arbitrarily chosen but must be stated and held constant.

time interval: The duration between two moments: $\Delta t = t_f - t_i$; always positive in classical physics.

motion diagram: A representation of an object's position at equally spaced time intervals, shown as a sequence of dots along the direction of motion.

position-time graph: A graph of position (y-axis) versus time (x-axis); the slope at any point equals the velocity.

uniform motion: Motion at constant velocity: equal displacements in equal time intervals; produces a straight line on an x-t graph.

changing motion: Motion in which velocity varies; produces a curved line on an x-t graph and unequal dot spacing in a motion diagram.

Concept Review Questions

Answer in complete sentences. No calculation required.

1. What does it mean to say that motion is relative? Give an example where an object is simultaneously at rest and moving, depending on the reference frame.
2. What is the difference between distance and displacement? Give an example where they have the same value, and one where they differ.
3. Why must you always state a reference point and positive direction before describing motion? What goes wrong if you don't?
4. What does a horizontal line on a position-time graph tell you about the object's motion?
5. What does a negative slope on a position-time graph mean physically?
6. Explain the difference between a steep straight line and a curve on a position-time graph. What does each tell you about acceleration?
7. Describe what a motion diagram with decreasing dot spacing tells you about the object's motion.
8. An athlete runs four laps of a 400 m track and returns to the start. What is the total distance? What is the displacement?
9. What is the difference between the time t and the time interval Δt? Give a physical example of each.
10. List the four representations of motion from Section 6.12. What is one advantage of each?

Motion Diagram Exercises

For each description, sketch a motion diagram (a sequence of dots) showing the motion. Label the time interval between dots and indicate the positive direction with an arrow.

11. A car travels at constant speed to the right for 6 seconds.
12. A ball starts slowly and speeds up as it rolls down a ramp for 5 seconds.
13. A runner decelerates from top speed and comes to a stop over 4 seconds.

14. A toy car moves at constant speed for 3 seconds, stops for 2 seconds, then moves slowly in the opposite direction for 3 seconds.
15. A ball is thrown upward (positive direction) and rises for 2 seconds then falls back down for 2 seconds.
16. For this motion diagram (equally spaced dots for 3 intervals, then gaps that increase for 3 more intervals), write a one-sentence description of the motion.

Position-Time Graph Practice

Use the following x-t graph description to answer questions 1–3. The graph shows four segments: (A) a straight line with positive slope from t=0 to t=3 s; (B) a horizontal line from t=3 s to t=5 s; (C) a straight line with negative slope from t=5 s to t=9 s; (D) a curved line (slope increasing) from t=9 s to t=12 s.

17. During which segment(s) is the object moving in the positive direction? Negative direction? At rest?
18. During which segment is the object moving fastest? Slowest (excluding rest)?
19. During which segment is the object accelerating (velocity changing)?
20. Sketch an x-t graph for the following motion: An object starts at x = +10 m, moves to x = +4 m over 3 seconds at constant speed, then rests for 2 seconds, then moves to x = +14 m over 4 seconds at constant speed.
21. An x-t graph shows a straight line passing through the points (0 s, 2.0 m) and (5.0 s, 12.0 m). (a) What is the slope? (b) What are the units of the slope? (c) What physical quantity does the slope represent?
22. Two objects A and B are shown on the same x-t graph. Object A starts at x = 0 and moves with a steep positive slope. Object B starts at x = +20 m and moves with a shallow negative slope. At what kind of point do their graphs intersect, and what does the intersection mean physically?

Distance and Displacement Problems

Show all work. Report displacement with sign and direction; report distance as a positive number.

23. A person walks 8.0 m east, then 5.0 m west. (a) What is the total distance traveled? (b) What is the displacement from start to finish?
24. A ball is thrown upward and rises 12.0 m before falling back to the thrower's hand. (a) What is the total distance traveled? (b) What is the displacement?
25. A car starts at position $x = +15.0$ m and drives to position $x = -8.0$ m. (a) What is the displacement? (b) What is the distance traveled (assuming no reversal)?
26. A hiker walks: 3.0 km north, then 4.0 km east, then 3.0 km south. (a) What is the total distance? (b) What is the magnitude of the displacement? (c) What is the direction of the displacement?
27. An ant crawls along the edge of a square tile of side 10 cm, completing three sides before stopping. (a) How far did it travel? (b) What is its displacement from start to finish?
28. A train starts at station A (defined as $x = 0$). It travels to station B at $x = +42.0$ km, then reverses to station C at $x = +18.0$ km. (a) Distance traveled? (b) Displacement from A to C? (c) Displacement from B to C?

Applied Motion Problems

These problems combine position, displacement, reference frames, and graph interpretation.

29. Two students watch the same ball roll across a table. Student A sits at one end and defines the positive direction as away from her. Student B sits at the other end and defines the positive direction as away from him. The ball rolls from A toward B by 1.5 m. (a) What is the displacement according to Student A? (b) What is the displacement according to Student B? (c) Do they disagree about the physics? Explain.
30. A dog runs 30 m due east in 6.0 s, then 20 m due west in 4.0 s. (a) What is the total distance? (b) What is the total displacement? (c) Draw a position-time graph of the dog's motion (sketch, not precise scale).
31. A position-time graph shows a straight line from (0 s, −4.0 m) to (8.0 s, +12.0 m). (a) What is the displacement? (b) What is the slope? (c) At what time does the object pass through $x = 0$?
32. An object's position is recorded every 2 seconds: x = 0, 4, 8, 12, 8, 4, 0 m (at t = 0, 2, 4, 6, 8, 10, 12 s). (a) Describe the motion in words. (b) What is the total distance traveled? (c) What is the displacement from $t = 0$ to $t = 12$ s?

33. A passenger on a train moving at 25 m/s eastward throws a ball at 5 m/s toward the front of the train. (a) What is the ball's velocity relative to the train? (b) Relative to the ground? (c) If the passenger throws the ball toward the back instead, what is the ball's velocity relative to the ground?

Chapter 6 Checkpoint

Before moving to Chapter 7, confirm you can do each of the following.

I can...	Section
Define motion as a change in position relative to a reference point	6.1
Calculate displacement as $\Delta x = x_f - x_i$ with correct sign	6.2
Distinguish distance (scalar, path-dependent) from displacement (vector, path-independent)	6.3
Identify the reference point and positive direction in any problem	6.4
Describe what dot spacing in a motion diagram reveals about speed	6.9
Read an x-t graph: identify rest, constant velocity, speeding up, slowing down	6.10
State that slope of an x-t graph equals velocity	6.10
Distinguish uniform motion from changing motion using graphs and diagrams	6.11
Translate a motion description between words, table, graph, and diagram	6.12
Identify and correct the six most common motion-description mistakes	6.13

Continue to Chapter 7: Speed, Velocity, and Acceleration →